XI Std Zoology Chapter-6 EM.indd 110 30/12/2022 10:42:06

Concept map

Mechanism of Respiration

- Inspiration and Expiration happens due to pressure changes in the thoracic cavity.
- By the contraction and expansion of diaphragm, external intercostal muscles and internal intercostal muscles the volume of the thoracic cavity is reduced or increased.
- The act of breathing is performed by expansion and contraction of the thoracic cavity.
- **INSPIRATION**
  - An active process by which fresh air is drawn into the lungs.
  - The contraction of external intercostal muscle causes the ribs to move anteriorly and outwardly.
  - The contraction of the diaphragm leads to flattening of inelastic, dome shaped central part of the diaphragm.
- **EXPIRATION**
  - A passive process by which air is exhaled from the lungs.
  - The diaphragm relaxes and rises to resume its original dome shape.
  - The ribs take their original position as a result of contraction of the internal intercostal muscles.

